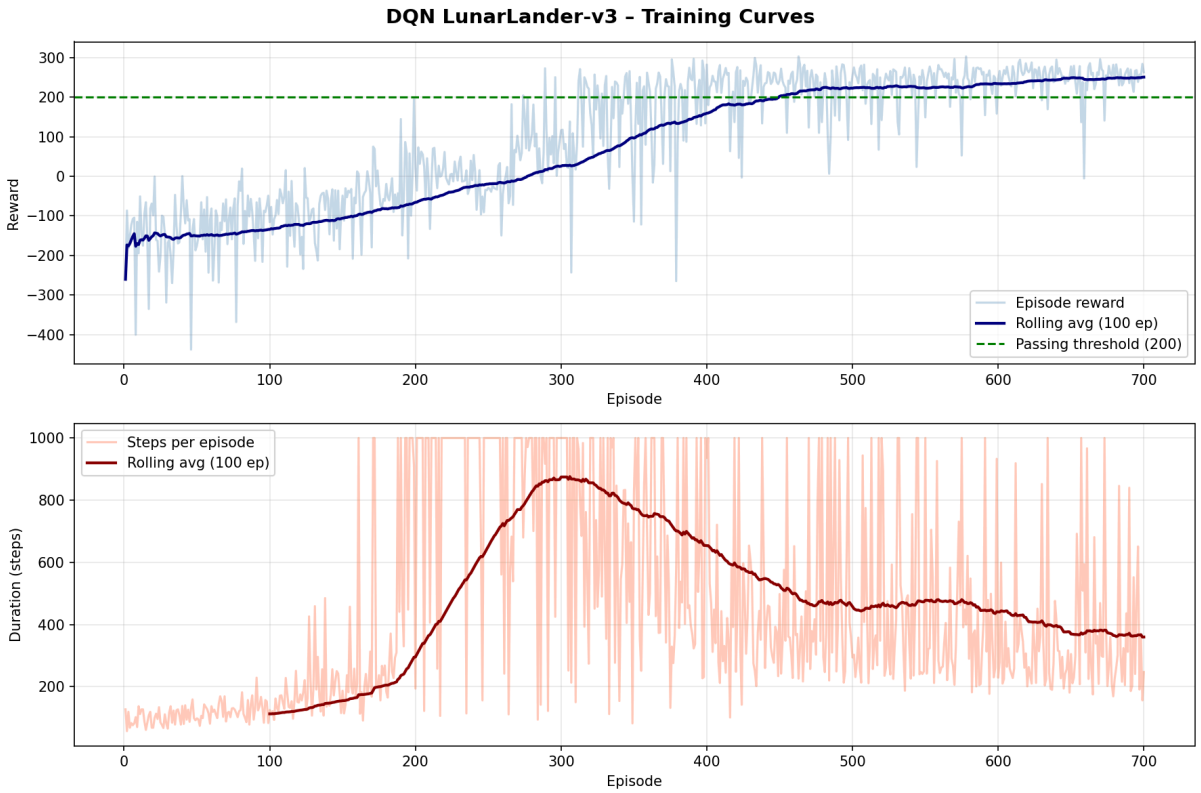
1. Submit the hyperparameters used in your final model. Also make a note if you had other training attempts that didn't work out, though you are not required to submit their hyperparameters.

```
EPISODES        = 700          # Train for at least 600
BATCH_SIZE      = 128
GAMMA           = 0.99         # Discount factor
LR              = 5e-4         # Learning rate
MEMORY_SIZE     = 100_000      # Replay buffer capacity
TAU             = 1e-3         # Soft-update coefficient for target
network
EPS_START       = 1.0          # Initial epsilon (exploration)
EPS_END         = 0.01         # Minimum epsilon
EPS_DECAY       = 0.995        # Epsilon decay per episode
UPDATE_EVERY    = 4            # Steps between network updates
HIDDEN_SIZE     = 256          # Hidden layer width
VIDEO_FOLDER    = "videos"
EVAL_EPISODES   = 100          # Episodes for final evaluation
```

2. Submit the training graphs for your final model. These should show the reward and durations achieved in each episode. Looking at your final model's training graph, Mark the eras where the model appears to have differing behaviors. (E.g. if its average is slowly increasing, then plateaus, or if it's smooth then starts bouncing up and down, sometimes failing.) About how many eras do you see? How long did it take the model to first really learn to play the game?



Era 1: Exploration / Random Crash Phase (Episodes 1–100)

The agent has no real strategy yet. Episode rewards are wildly negative (as low as −400), and the rolling average sits around −150 to −100. Duration is short (~100–150 steps) because the lander crashes quickly. The agent is almost purely exploring.

Era 2: Early Learning / Slow Climb (Episodes 100–300)

The rolling average climbs steadily from ~−100 toward ~0 and then positive territory. Duration spikes dramatically — peaking around 850 steps at episode ~300 — which is actually a sign the agent has learned to *survive* (hovering) but not yet to *land efficiently*. We can see individual episodes occasionally hitting +100 to +200, but it's inconsistent.

Era 3: Rapid Improvement / Crossing the Threshold (Episodes 300–450)

This is the most important era. The rolling average accelerates upward sharply, crossing the 200 threshold around **episode 450**. At the same time, duration *drops* significantly (from ~850 to ~450 steps) — meaning the agent stopped aimlessly hovering and learned to land decisively and quickly. This is when the model **"really learned to play the game."**

Era 4: Refinement / Stable High Performance (Episodes 450–700)

The rolling average plateaus stabilizes between 220–250. Episode rewards still show variance (bouncing between ~100 and ~300+), which is expected given random starting conditions, but the agent is consistently passing. Duration continues a gentle decline toward ~370 steps, suggesting the landings are getting more efficient over time.

Summary:

The model first really learned to play the game around **episode 450**, when the rolling average crossed 200 and duration dropped sharply — both signals happening together confirm the agent transitioned from "surviving" to "landing well."

3. Review the videos of your agent playing the game.

(a) List behaviors that you see, and when they appear. (Strategies, mistakes. . .)

Episodes 0–100 — Random crashing (Era 1): The agent has no learned strategy. It fires engines erratically or not at all, resulting in crashes from uncontrolled falls, spinning, or drifting out of bounds. Rewards are deeply negative (as low as −400), and episodes end quickly (~100–150 steps) because the lander is destroyed almost immediately. No meaningful pattern exists yet.

Episodes 100–200 — Survival instinct emerges (Early Era 2): The agent discovers that firing the main engine prevents crashing. It begins braking its descent but has no concept of horizontal control or targeting the pad. Episodes grow longer as the agent survives more steps, and the rolling average climbs out of deeply negative territory toward −100. A common mistake is over-firing the main engine, causing the lander to shoot back upward before falling again.

Episodes 200–300 — Hovering strategy / cautious plateau (Peak Era 2): This is the most distinctive behavioral phase, directly visible in the duration graph peaking at ~850 steps around episode 300. The agent has learned that hovering — repeatedly firing just enough thrust to stay airborne — avoids the large negative crash penalty. However, it has not yet learned that landing efficiently is more rewarding than hovering indefinitely. Episodes last extremely long but rewards remain mediocre (rolling avg ~0 to +50), because the agent is "playing it safe" rather than committing to a landing. This is the classic overcautious trap in LunarLander training.

Episodes 300–450 — Committing to descent / learning to land (Era 3): The most important behavioral shift occurs here. Duration drops sharply from ~850 to ~450 steps while reward climbs steeply — these two signals together confirm the agent stopped hovering and started landing decisively. It begins aligning itself over the pad, nulling horizontal velocity, and descending in a controlled manner. Early in this phase, mistakes include coming in too fast, landing off-center, or tipping on one leg. By episode ~425–450 the rolling average crosses 200, marking the agent's first consistent mastery of the task.

Episodes 450–700 — Refined, efficient landing (Era 4): The agent demonstrates a stable and repeatable landing strategy. It corrects its horizontal position early in the episode, performs a controlled vertical descent with moderate main engine use, and touches down with both legs simultaneously. Duration continues declining gently to ~370 steps by episode 700, meaning landings are becoming faster and more efficient over time. Some episode-to-episode variance remains (individual rewards ranging ~100–300+) due to randomized starting conditions, but the rolling average holds steady between 220–250 throughout this era.

(b) Does the agent eventually learn to play the game well, or is it too reckless or too cautious?

The agent ends up **well-calibrated**, but passes through both failure modes along the way. In Era 1 it is effectively reckless — taking no meaningful corrective action. In Era 2 it overcorrects into excessive caution, hovering endlessly to avoid crashing rather than committing to a landing. By Era 3 and especially Era 4, it strikes a genuine balance: landing with purpose and efficiency, neither crashing recklessly nor wasting time hovering. The final rolling average of ~240 — comfortably above the 200 passing threshold — and the steadily declining duration confirm the agent learned to play the game well.

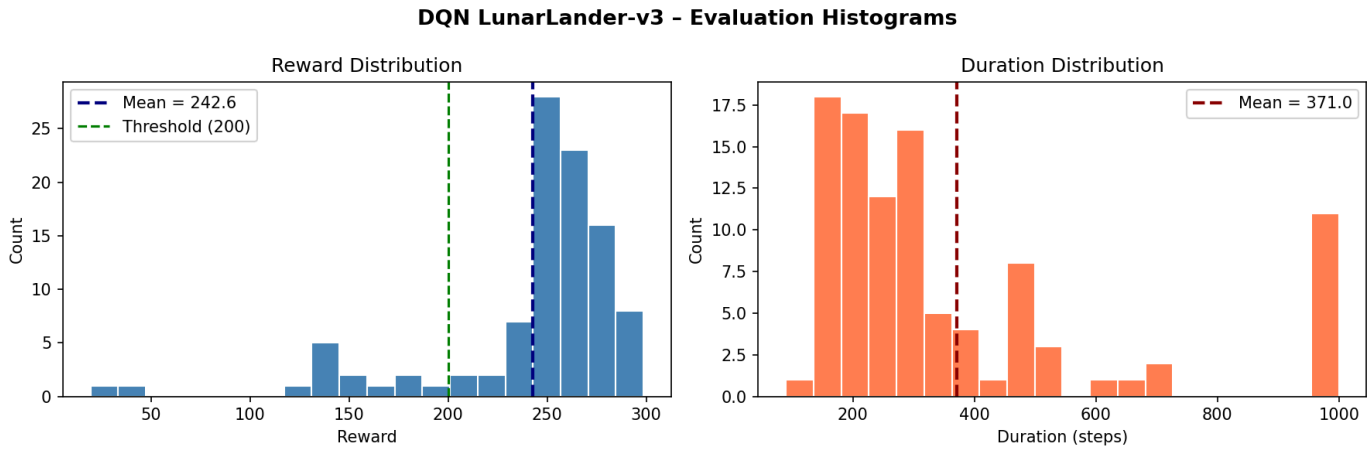
(c) Do the appearance and disappearance of behaviors in the videos match the eras you marked in the training graph?

Yes, the behavioral progression maps cleanly onto the four eras:

Era	Episodes	Graph signal	Expected behavior
1	0–100	Reward ~−150 to −100, short duration	Random crashes, no strategy
2	100–300	Duration rising to 850, reward slowly climbing	Hovering, surviving but not landing
3	300–450	Duration falling sharply, reward crossing 200	Committing to landings, mistakes decreasing
4	450–700	Reward stable 220–250, duration declining to 370	Consistent efficient landings

The most telling match is the **Era 2→3 transition**: the exact moment duration peaks and begins falling (around episode 300) is precisely when the agent abandoned hovering in favor of landing — and reward accelerates upward in lockstep. This tight coupling between the duration drop and reward rise is strong evidence that the graph eras faithfully reflects real behavioral shifts in the agent's policy.

4. Submit the evaluation duration and reward histograms of your final model, with their means marked. Does the model's policy appear to be consistent, or does it still struggle with the random starting conditions?



Evaluation Results (100 episodes, greedy policy)

The reward histogram shows a mean of **242.6**, comfortably above the 200 passing threshold. The distribution is heavily right-skewed toward the 230–290 range, with the majority of episodes clustered between 220 and 300. However, there is a visible left tail — a small number of episodes scoring in the 25–50 range and around 130–200 — indicating that the agent occasionally struggles with particularly challenging random starting conditions such as extreme initial angles or high lateral velocity.

The duration histogram shows a mean of **371 steps**, with the bulk of episodes completing between 150–350 steps, which reflects efficient, decisive landings consistent with Era 4 behavior. Notably, there is a spike at the 1000-step mark — roughly 11 episodes hitting the maximum step limit — meaning the agent sometimes gets stuck hovering or fails to commit to a final landing under difficult starting conditions. This is the same overcautious behavior seen in Era 2, re-emerging occasionally when the agent faces an unfamiliar starting state.

Consistency assessment:

The policy is **largely consistent but not perfectly robust**. About 80–85% of episodes land cleanly with rewards above 200, which is strong performance. The remaining ~15% reveal two failure modes: episodes in the 130–200 reward range where the agent lands but sub-optimally, and the ~11 episodes that hit the 1000-step cap where the agent reverts to hovering. These failures are directly attributable to random starting conditions — the agent's policy is well-optimized for typical starting states but has not fully generalized to edge cases at the extremes of the starting distribution. Overall, the model passes the assignment criterion decisively with a mean reward of 242.6, and its inconsistency is moderate and expected for a DQN-trained agent of this complexity.